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4 May 1972
SUPERSEDING
MIL-T-45508A(WC)
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MILITARY SPECIFICATION

TARGET HOLDING MECHANISM, TRAINFIRE, M31A1

1. SCOPE

1.1 This specification covers one type of motor driven Target Holding Mechanism (Trainfire). It is designed to operate on 120 volt, 60 cycles, alternating current (AC) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-6872 - Soldering Process, General Specification for
MIL-S-12332 - Welding, Resistance, Spot and Projection; for Fabricating Assemblies of Low-Carbon Steel
MIL-P-14232 - Parts, Equipment and Tools for Ordnance Materiels, Packaging of

STANDARDS

Federal

FED-STD-595 - Colors

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of U. S. Military Property
MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-1262 - Welding Procedures for Constructional Steels

FSC 6920

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DRAWINGS

Ordnance Corps

C8410714 - Switch
C8426438 - Target, Kneeling
F8421550 - Box Assembly, Operating
F8421679 - Target Holding Mechanism, Trainfire, M31A1

PUBLICATIONS

Ordnance Corps

P8421679 - Packaging Data Sheet, for Mechanism, Target Holding, Trainfire M31A1

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI) Standards

ANSI Y14.5 - Dimensioning and Tolerancing

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

American Welding Society (AWS)

AWS A2.0-68 - Welding Symbols

(Application for copies should be addressed to the American Welding Society, Inc., 345 East 47th Street, New York, N.Y. 10017.)

3. REQUIREMENTS

3.1 Material. Material shall conform to the material specifications referenced on the drawings.

3.2 Dimensions. Dimensions of parts and assemblies shall conform to the applicable drawings of F8421679. Interpretation of dimensional and tolerance requirements shall be in accordance with ANSI Y14.5.

3.3 Constructions.

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3.3.1 Assembly. Parts shall be positioned and attached in accordance with the applicable drawings of Drawing F8421679. Approved source parts specified on the drawings shall meet with Army (Weapons Command) approval, prior to assembly.

3.3.2 Welding. Steel weldments of joints for the mechanism shall be in accordance with MIL-STD-1262. Spot welding shall be in accordance with MIL-W-12332. Welded joints and spot welds shall conform to weld symbols as interpreted by AWS A2.0-68. Actual welding shall not proceed prior to Government approval of the contractor's welding procedure and test specimens.

3.3.3 Soldering. The soldering of electrical connections in the mechanism shall be in accordance with MIL-S-6872.

3.4 Protective finishes.

3.4.1 Inorganic finishes. Inorganic finishes shall be as specified on the detailed drawings and in accordance with MIL-STD-171.

3.4.2 Painting. The Target Holding Mechanism and parts shall be painted as specified on the drawing and in accordance with MIL-STD-171 (see 6.3.2). Unless otherwise specified in the contract or order, the finish number of MIL-STD-171 to be used for cleaning shall be optional with the contractor.

3.4.2.1 Parts. The following parts shall not be primed or painted: interior of Operating Box (8421571) and finished surfaces of flange, including "O" ring groove, inside of Operating Box Cover (8421568), all parts inside the operating box assembly, and external and internal screw threads.

3.4.2.2 Assembled mechanism. All exposed surfaces of the assembled mechanism shall be painted, except rubber and plastic parts and bearing surfaces. The exposed end only of the motor shaft shall be painted.

3.5 Performance and product characteristics.

3.5.1 Time cycle. The time required to raise Target C8426438 shall not exceed 1.8 seconds. The time required to lower the same target shall not exceed 1.5 seconds at room temperature of 75°F, plus or minus 10°F. (See 6.3.3).

3.5.2 Operation. The impact upon the raised Target C8426438 of a one ounce, minus 0.2 ounce, spherical weight, traveling through a 90°, minus 15° arc, on a 20 inch, plus or minus one inch radius, shall actuate the switches C8410714 and kill the target (cause it to fall).

3.5.3 Waterproofness. The Box Assembly, Operating, F8421550, with an air pressure of 2.0 pounds per square inch (psi), plus 0.1 psi, applied shall show no evidence of leakage when the mechanism is submerged in water

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to a depth of one inch, plus one inch, above the top of the box assembly, for a period of one minute.

3.5.4 Reliability. The target mechanism with Target C8426438 in place shall be capable of operation for 240 cycles at a rate of 4 cycles per minute at temperatures of 0°F, plus or minus 5°F, and at 125°F, plus or minus 5°F. At a temperature of 75°F, plus or minus 10°F, the target mechanism shall be capable of continuous operation for 9600 cycles.

3.6 Marking. Marking of the target mechanism and component parts shall be in accordance with the applicable drawings and MIL-STD-130. Serial numbers for the identification plates will be furnished by the procuring activity (see 6.3.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preliminary inspection. Prior to initiating acceptance inspections, two completed target holding mechanisms, taken at random from the initial lot, shall be subjected to the reliability test of 4.5.4 to determine compliance with 3.5.4. If either of the two target holding mechanisms fails to pass this test, acceptance inspection shall be deferred pending corrective action by the contractor. After Government verification of corrective action by the contractor, the test shall be repeated with two target holding mechanisms other than the two originally tested. Acceptance inspection shall be initiated after two mechanisms have passed this test. The disposition of the target holding mechanisms used in this test shall be determined by the Contracting Officer. Unless otherwise specified in the contract or order, the test of 4.5.4 shall be repeated on one target holding mechanism taken at random from each 1000 target holding mechanisms produced, or as determined by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 Lot formation. Unless otherwise specified in the contract or order, a lot shall consist of all items manufactured or packaged under essentially the same conditions.

4.3.2 Sampling.

4.3.2.1 Sampling for examination. A random sample of target holding mechanisms shall be taken from each lot in accordance with MIL-STD-105, at Inspection Level II. Acceptable quality levels (AQL) shall be as specified in Table I.

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4.3.2.2 Sampling for packaging. Sampling for examination of packaging, packing, and marking shall be in accordance with MIL-P-14232.

4.3.3 Certification.

4.3.3.1 For parts covered by a procurement specification. When parts are covered by a procurement specification referenced on the drawing, the Government shall be furnished certification by the contractor that the parts are in compliance with the requirements of the applicable drawing and referenced specification.

4.3.3.2 Material. The contractor shall furnish certification that materials used in construction of the target holding mechanism conform to the material specification referenced on the applicable drawings.

4.3.3.3 Protective coatings. The Government shall be furnished certification by the contractor that cleaning, chemical finishes and painting conform to the finish and system numbers of MIL-STD-171 (see 3.4).

4.3.3.4 For Ordnance and Military standards (Commercial). The Government shall be furnished certification by the contractor that the commercial parts furnished are in compliance with the requirements of the applicable standards and drawings.

4.4 Examination.

4.4.1 Each of the target holding mechanisms taken in accordance with 4.3.2.1, shall be examined to verify compliance with this specification. Examination shall be conducted as specified in Table 1.

TABLE 1 - CLASSIFICATION OF DEFECTS

Categories	Defects	Inspection Method
Critical	None defined	
Major	AQL = 1.0 percent	
101	Mechanism not complete, component parts missing	Visual
102	Missing welds, or visible welding defects	Visual
103	Visible soldering defects	Visual
Minor	AQL = 4.0 percent	
201	Defects of painting, including runs, blisters, sags, incomplete coverage or wrong color	Visual
202	Identification or instructional marking incomplete or illegible	Visual

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4.4 Packaging, packing and marking. Examination of packaging, packing and marking shall be performed in accordance with MIL-P-14232.

4.5 Test procedures.

4.5.1 Waterproofness. Each Box Assembly, Operating, F8421550 shall be subjected to the following test. Any box assembly failing this test shall be cause for corrective action by the contractor. Remove power input plug and assemble all receptacle caps securely in place. Replace pipe plug with air hose connection and supply air pressure of 0.4, plus 0.1 psi. With air pressure maintained, the box assembly shall be submerged in water of temperatures of 75°F, plus or minus 10°F, to a depth so that the water level is 1.0 inch, plus 1.0 inch, above the box assembly. Increase the air pressure to 2.0 psi, plus 0.1 psi, and maintain for a minimum period of one minute. There shall be no evidence of leaks as indicated by air bubbles.

(CAUTION) Prior to raising the box assembly from submerged condition, decrease the air pressure to 0.4 psi maximum to prevent damage to the bellows.

4.5.2 Time cycle. Each target holding mechanism shall be subjected to the following test. Any mechanism failing this test shall be cause for corrective action by the contractor. Prior to test, the box shall show continuity to pin "C" of the input receptacle (8421648), for proper grounding. At a room temperature of 75°F, plus or minus 10°F, the target holding mechanism, with Target C8426438 attached and mount assembly in extended position, shall be connected to a 120 volt, 60 cycle, AC power supply. The target shall be actuated by means of a control switch connected to the external binding posts, U, C, and D, a minimum of ten complete cycles. An appropriate timing device shall be used to measure the time cycle for compliance with 3.5.1. A 12-16 volt, 0.2 ampere lamp shall be connected across the external binding posts L and C. The lamp shall light when the control switch is thrown to "target up" position and shall go out when the control switch is thrown to "target down" position.

4.5.3 Operation. Each target holding mechanism shall be subjected to the following test. Any mechanism failing this test shall be cause for corrective action by the contractor. Target C8426438 shall be in "up" position and the target holding mechanism shall be connected to a 120 volt, 60 cycle, AC power supply. The apparatus to be used for imparting the impact to the target shall be a pendulum type. A spherical shaped weight of one ounce, minus 0.2 ounce, attached to a suitable cord shall travel thru an arc of 90°, minus 15°, before impact, on a radius of 20, plus or minus 1 inch. The point of impact of the weight on the target shall be at the approximate center of the target.

4.5.4 Reliability test.

a. The target holding mechanisms of 4.2 shall be subjected to a 0°F, plus or minus 5°F, temperature, in a cold room, for a minimum of

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six hours and then functioned 240 cycles (1 hour at 4 cycles per minute) while still in the cold room. A fifteen cycle warm-up will be permitted. The mechanism shall have Target C8426438 mounted in the holder while functioning.

b. The target holding mechanism shall then be subjected to 125°F, plus or minus 5°F, temperature in a hot room for a minimum of six hours and then functioned 240 cycles (1 hour at 4 cycles per minute) while still in the hot room. The mechanism shall have Target C8426438 mounted in the holder while functioning.

c. After successfully passing the above tests, the mechanism shall be operated continuously for 9600 cycles at a temperature of 75°F, plus or minus 10°F, at four cycles per minute. The mechanism shall have Target C8426438 mounted in the holder while functioning.

d. The movement of the target from the vertical position to the horizontal position and return to the vertical position shall constitute one cycle.

5. PREPARATION FOR DELIVERY.

5.1 Mechanism, Target Holding, M31A1. Preservation, packaging, packing, and marking shall be in accordance with Packaging Data Sheet P8421679.

5.2 Repair parts. Repair parts shall be preserved, packaged, packed, and marked as specified on the packaging data sheets furnished by the procuring activity.

5.3 Level of packaging and packing. The level of protection required shall be as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. The Target Holding Mechanism, Trainfire, M31A1, is for use with automatic or semi-automatic rifle fire. The M31A1 mechanism accomodates either the Kneeling Target C8426438 or the Prone Target C8426556. Target C8426438 only will be used for performance tests of the mechanism (see 6.3.3).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number and date of applicable drawings and packaging data sheet(s).
c. Finish number of MIL-STD-171 for cleaning surfaces prior to painting, if required.
d. System finish number for component parts, if other than that specified (3.4.2).

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e. Place of inspection and tests.
f. Level of packaging and packing required.
g. List of repair parts and packaging data sheets, as applicable.

6.3 Suggested contractual features.

6.3.1 Serial numbers. The contractor should be furnished a block of serial numbers by the Contracting Officer to be used for marking the M31A1 nameplates.

6.3.2 Colors. Color Chip 24087 of FED-STD-595 should be furnished the contractor to be used as a color guide in painting.

6.3.3 Government-furnished equipment. An adequate number of Targets (C8426438) should be furnished the contractor to determine compliance with 3.5.1, 3.5.2, and 3.5.4.

Custodian:
Army - WC

Reviewer:
Army - WC

User:
None

Preparing Activity:
Army - WC

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